

COMPASS

Two-Dimensional Semantic Dials Inside a Single Attention Head

*A rank-2 causal editing tool that reads, writes, and routes
per-domain bias.*

GPT-2 · Phi-3-mini · Gemma-2-2B · Llama-3.2-3B

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Pages 2–9: one poster panel per page, A3 landscape.

Page 10: take-home message.

Build: `pdflatex poster_panels.tex`

1. What is a compass?

OV-SVD $\rightarrow$ rank-2 dial

- For head h at layer ℓ , factor $W_{OV} = W_V W_O = U \Sigma V^\top$.

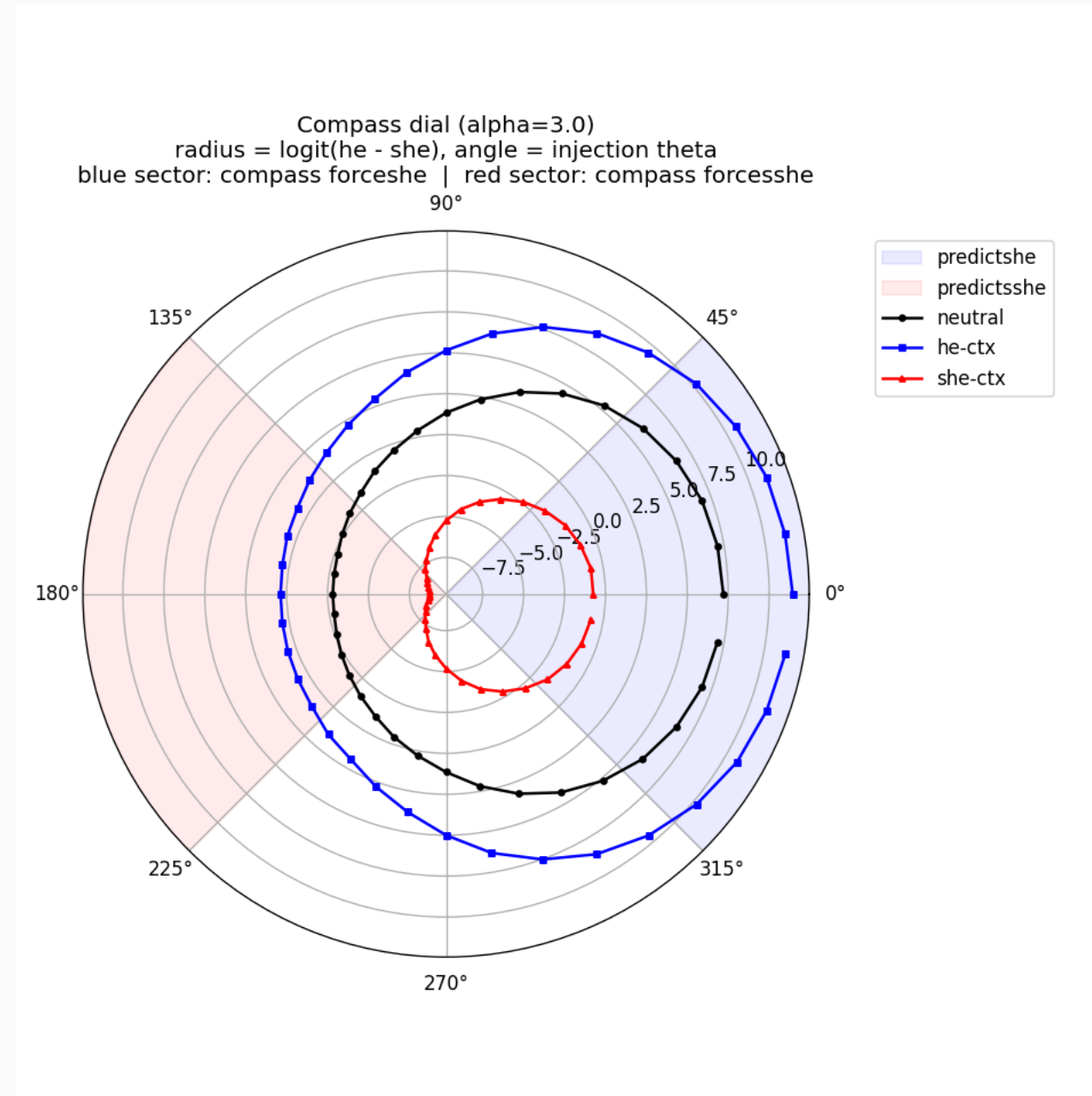
- A **compass plane** is two right-singular vectors (u_i, u_j) . Inject along it:

$$v(\theta, \alpha) = \alpha \sigma_i \cos \theta u_i + \alpha \sigma_j \sin \theta u_j.$$

- If the plane is causal, the logit-difference swings **sinusoidally** in θ and **linearly** in α ; if it is just correlational, random planes and label permutations reproduce the same numbers.
- The rest of this deck shows that (i) the sinusoidal + linear response really holds, (ii) compasses exist in every model we looked at, and (iii) they can drive a debiasing stack.

2. Sinusoidal response

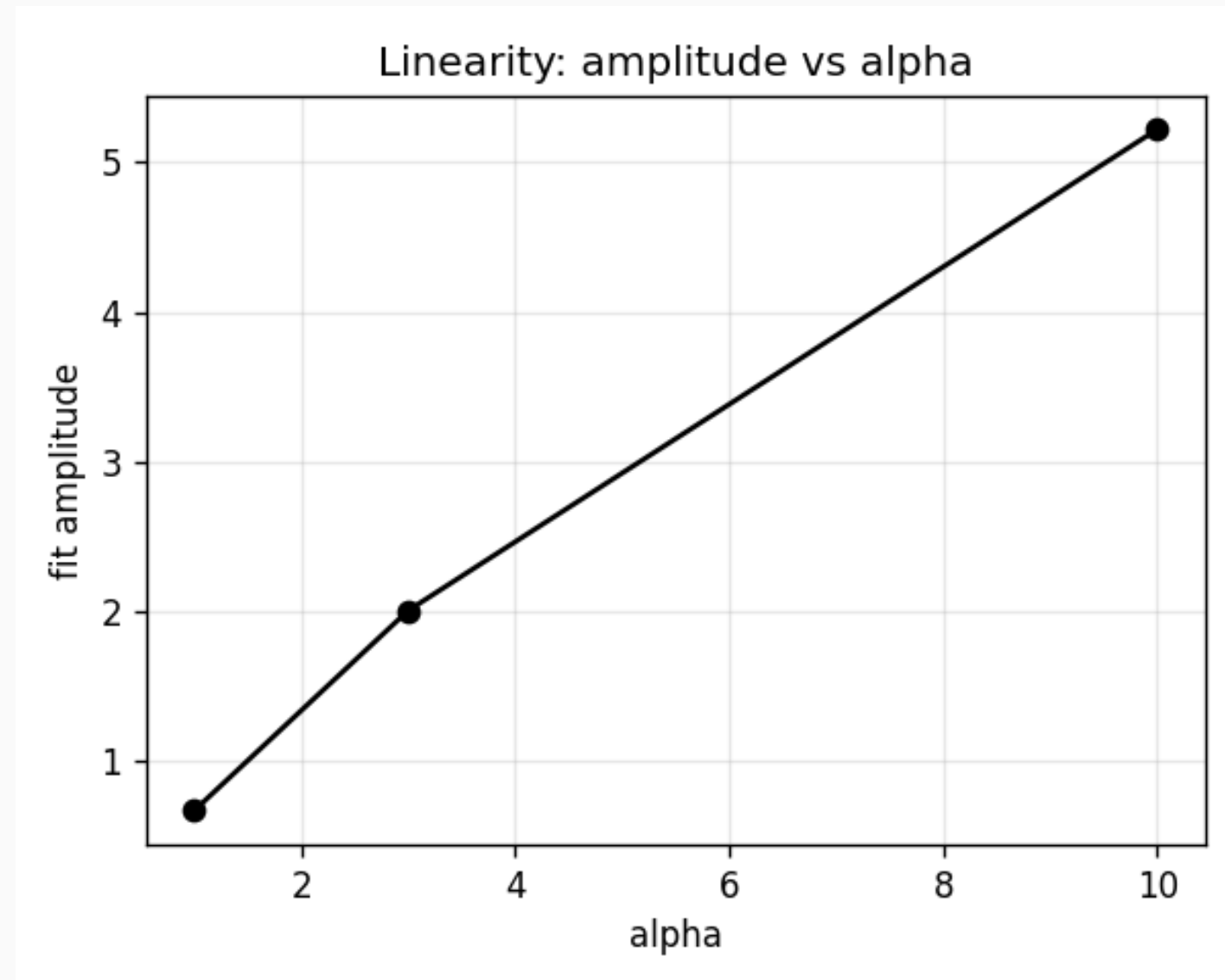
Phi-3 Mini L28H1 gender compass



Radius = $\text{logit}(he - she)$; angle = injection θ . Poles at 0°/180° decode to *he / his / him* vs. *she / her / hers*. Blue sector: *he*-context; red sector: *she*-context; black: neutral.

3. Linearity + phase stability

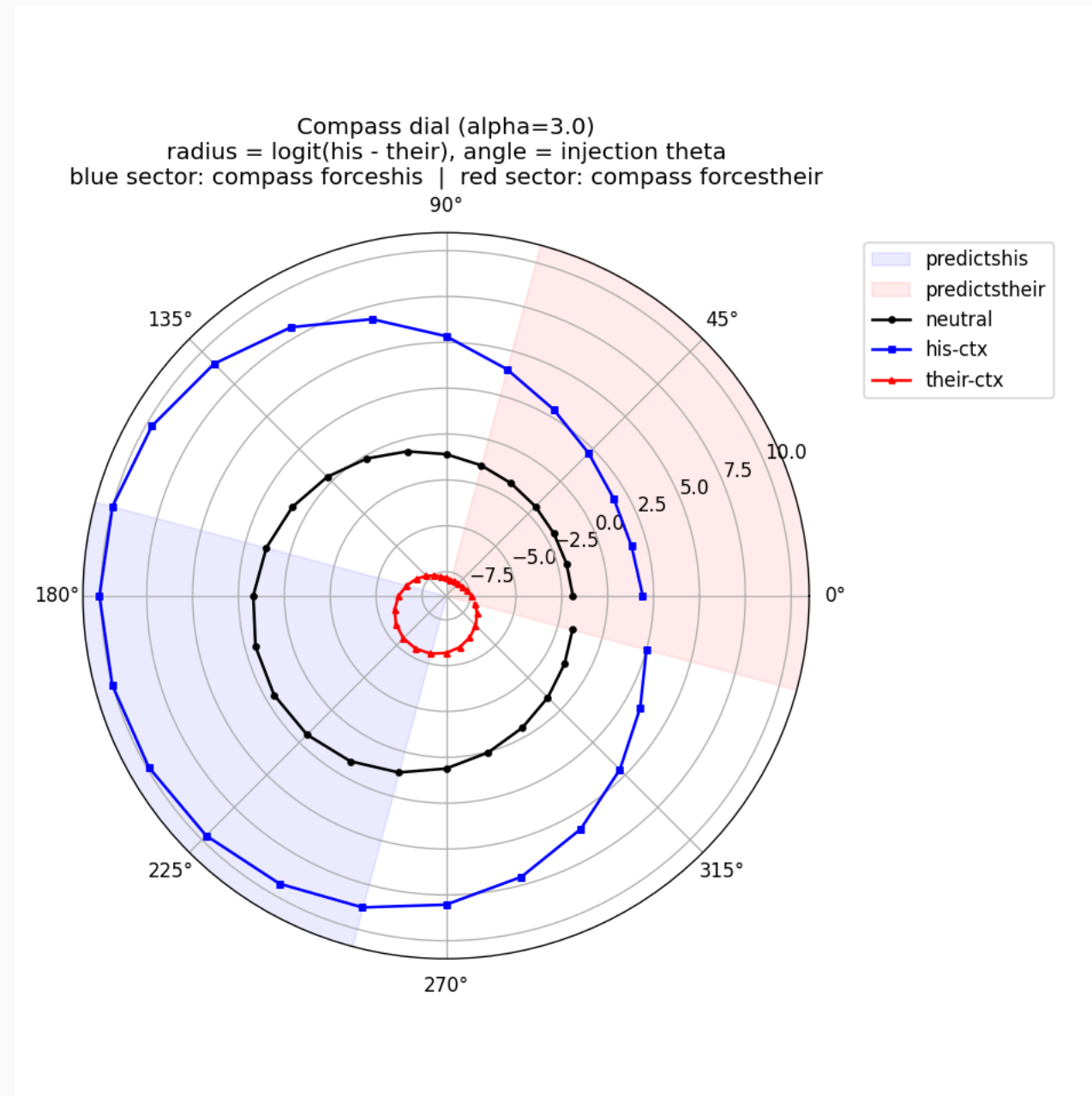
Two-pillar causal test



- Amplitude grows **linearly** with α ($R^2 \geq 0.993$).
- Phase drift $\leq 3^\circ$ across $\alpha \in \{1, 3, 10\}$.
- Random planes and permuted labels kill both signals — so the plane is not a geometric accident.

4. Not just gender

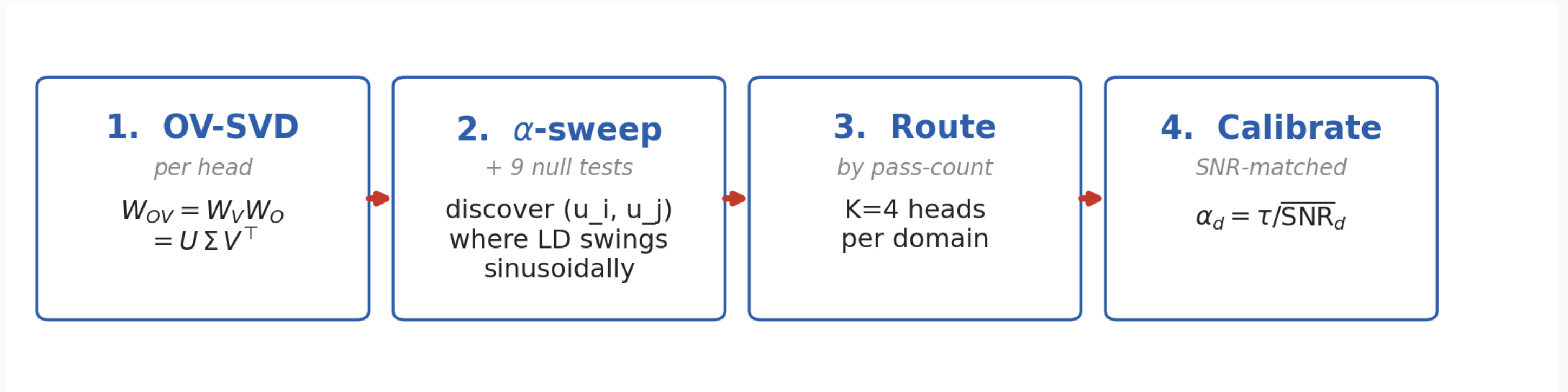
Same recipe, different axis



The same two-vector recipe finds *his* / *their* (number) and temporal (*was* / *is*) planes in the same heads. Compasses are **not** a pronoun-specific artifact — gender, race, number, tense, and entity all show up as rank-2 planes.

5. Pipeline

From raw weights to a routed debiasing dial



- **OV-SVD** per head → **α-sweep** + **9 null tests** → **route by pass-count** → **calibrate** $\alpha_d = \tau / \overline{\text{SNR}}_d$.
- Blind-scan pass rates vs. full-OV null (95% CI): GPT-2 2.06%, Phi-3 1.18%, Gemma 2.02%, Llama 0.22% — all significantly above the null.
- Nine null tests knock out correlational planes: permuted labels, random planes (1000 MC), full-OV ablate, semantic-ablate, principal-angles, causal-patch, self-test, decode-heads, cyclicity.

6. Four models, same structure

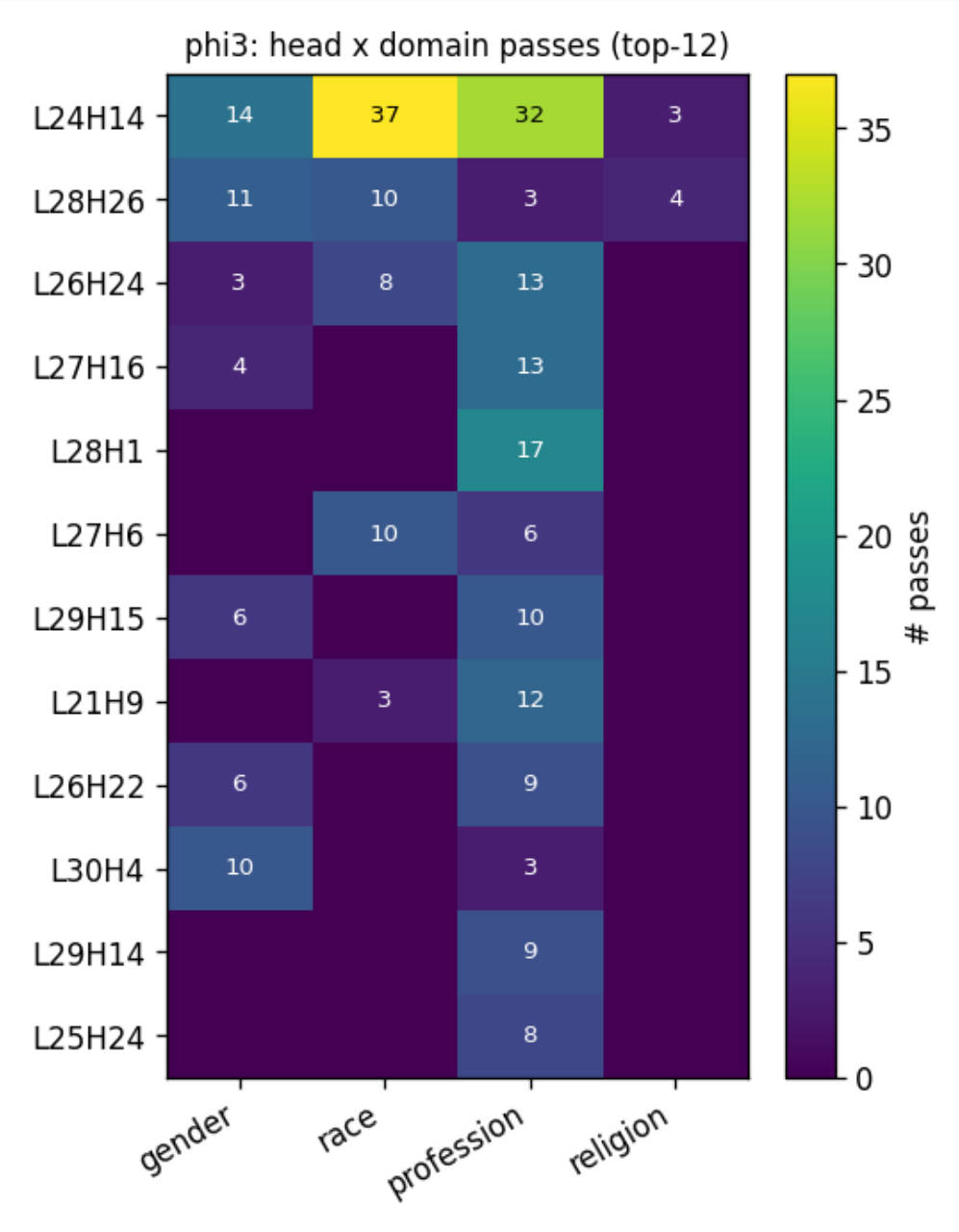
Gender compass summary

Model	head	$A(10)$	$\Delta\phi$
GPT-2 Small	L10H9	14.16	0.3°
Phi-3 Mini	L28H1	12.85	1.3°
Gemma-2-2B	L21H4	2.19	1.3°
Llama-3.2-3B	L26H14	3.01	2.4°

- Amplitudes differ by $\sim 5\times$ across families (GPT-2 $\rightarrow$ Gemma).
- Phase is stable everywhere — the dial is always sharp.
- Same two-pillar recipe discovers the plane independently of depth, width, vocab size.

7. Head–domain routing

Heads specialize by bias type

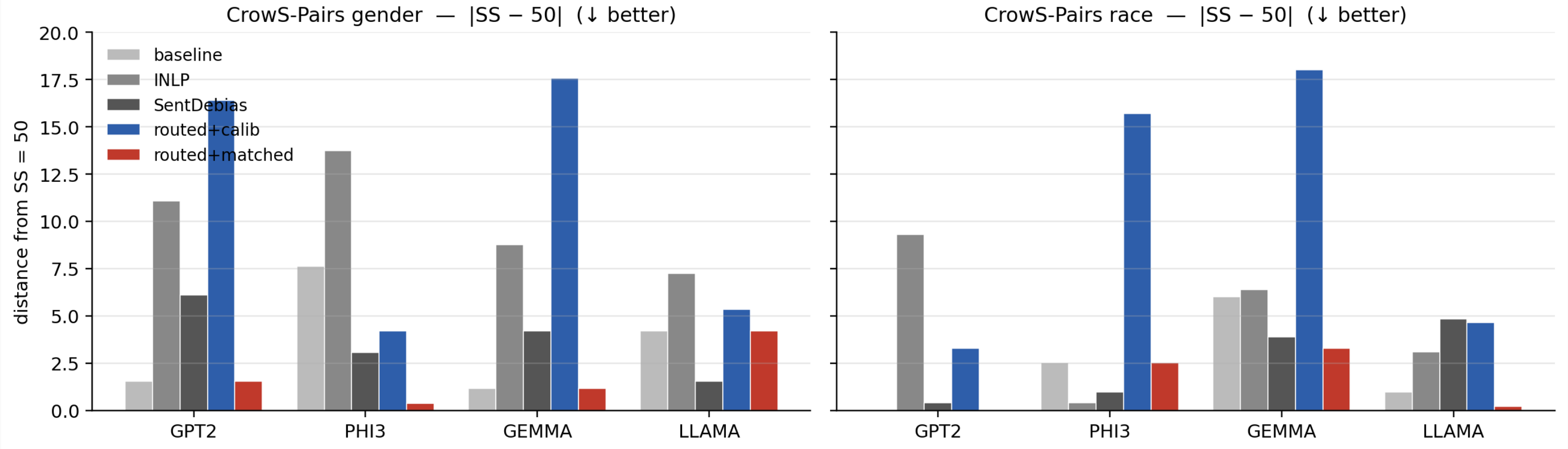


Each passing Phi-3 head concentrates on exactly one of {gender, race, profession, religion}. We inject only on rows whose `bias_type` matches, at an SNR-calibrated per-domain $\alpha_d = \tau / \overline{\text{SNR}}_d$ — one dial per axis, not one global scale.

8. Debiasing beats INLP / SentenceDebias

CrowS-Pairs $|SS - 50|$ (lower is better)

Routed + matched-strength compass beats INLP on 7/8 cells and SentenceDebias on 6/8



- Gender + race, 4 models $\times$ 5 methods.
- **routed + matched-strength** (red) beats **INLP** on 7/8 cells and **SentenceDebias** on 6/8.
- TruthfulQA MC1 moves by $\leq \pm 3$ pts — capability preserved.

Take-home

Three things to walk away with

1. Any attention head that writes a categorical axis contains a **2D compass plane** you can find by looking at *one SVD*.
 2. The plane is **causal, linear, and phase-stable** — not just a correlation.
 3. Routing + per-domain α turns these planes into a **debiasing dial** that matches or beats INLP / SentenceDebias while preserving capability.
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Code, scripts, and every CSV behind these tables:

<https://github.com/<your-account>/Beyond-Components>